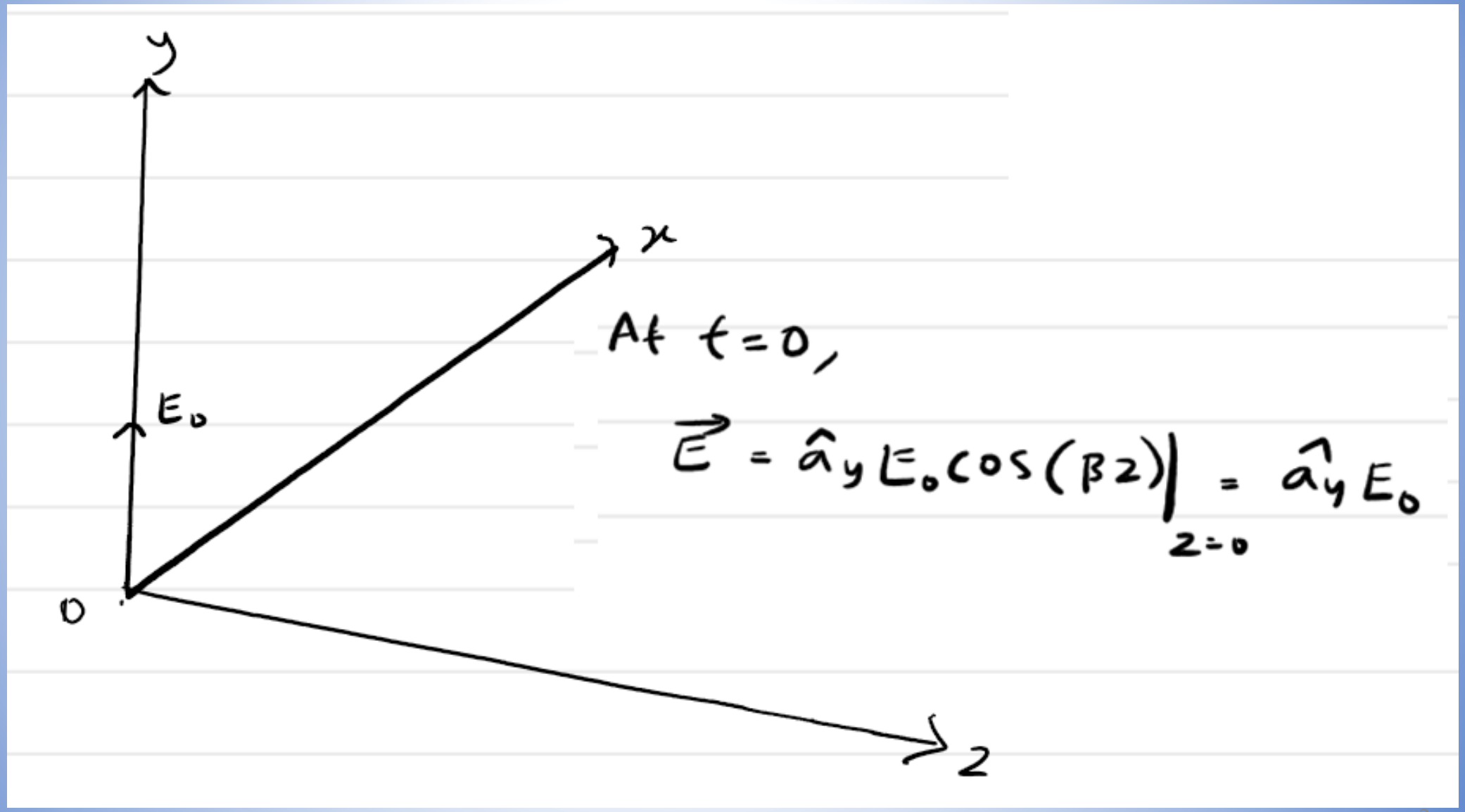
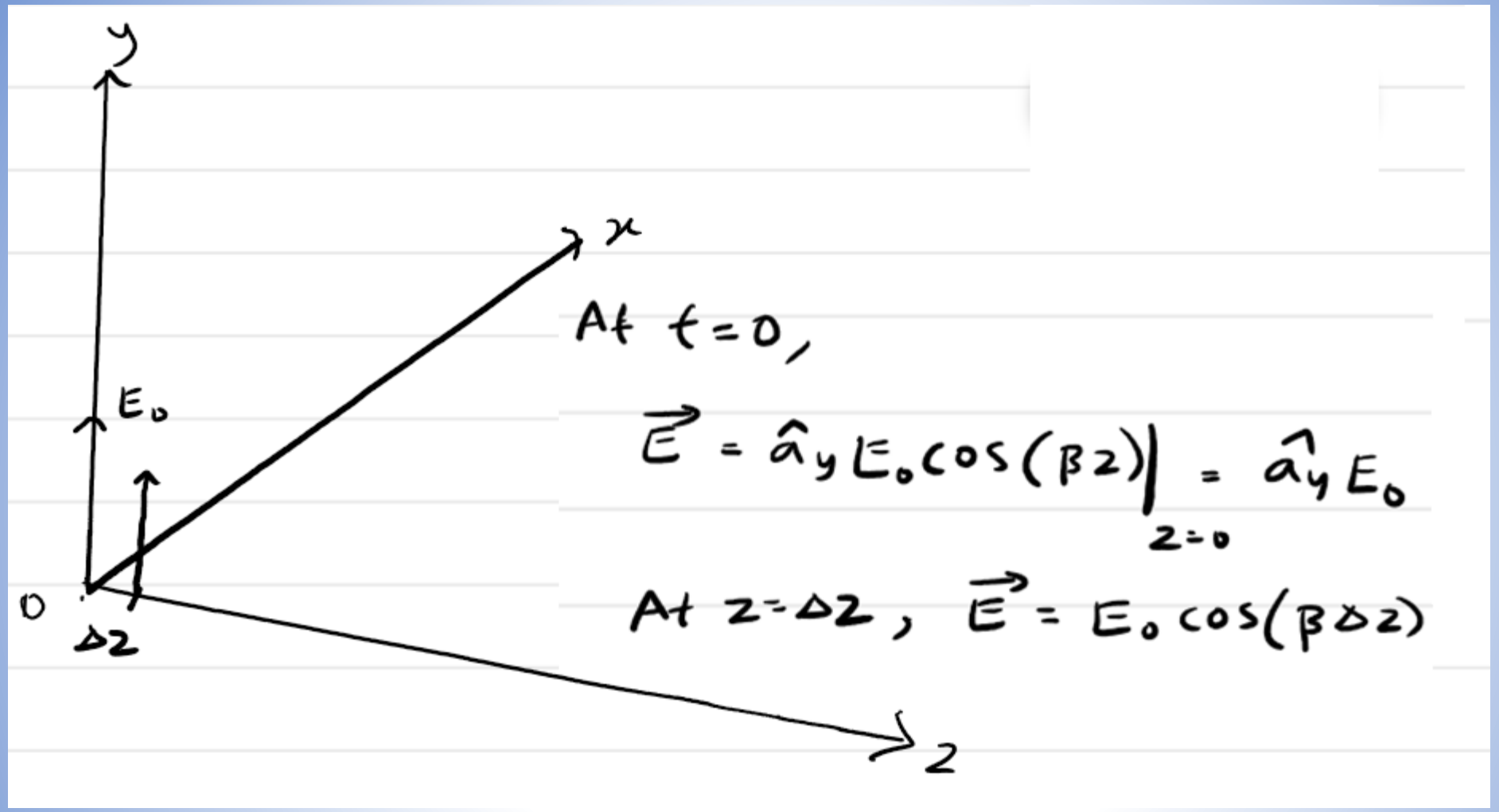
Uniform Plane Wave Propagation

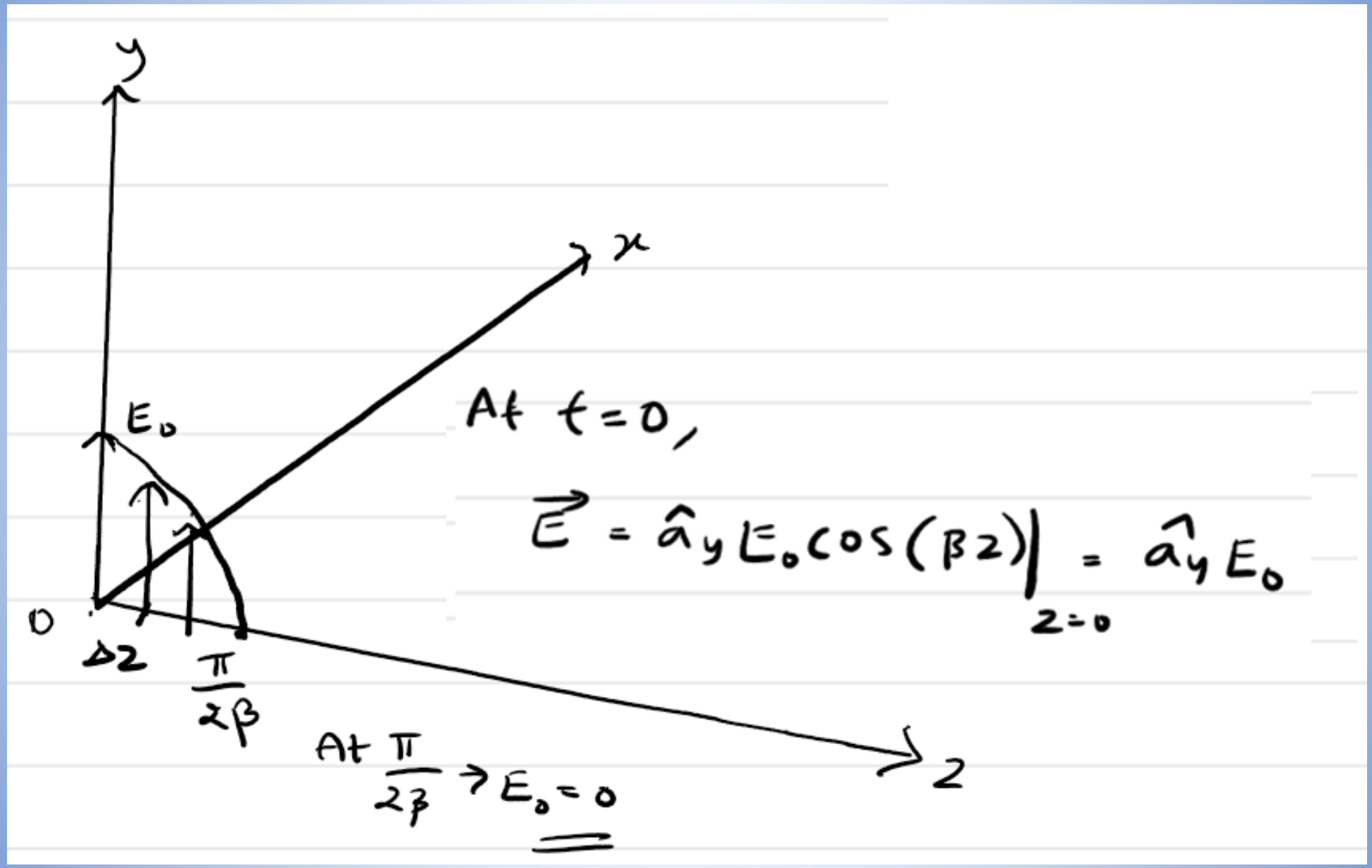
Dr. Sampath Kumar
Associate Professor
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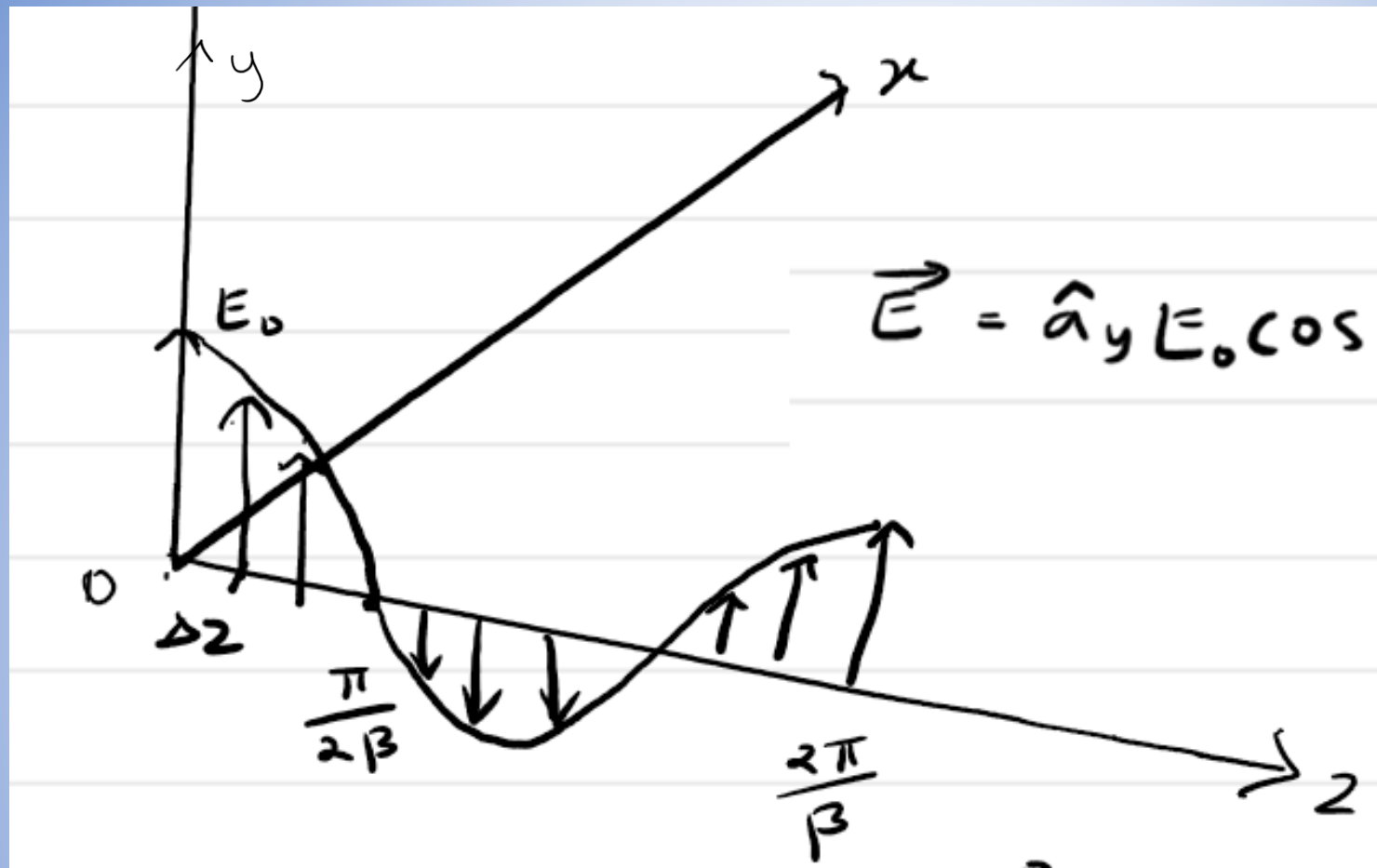
$$\vec{E} = \hat{a}_y E_o \cos(\omega t - \beta z)$$





Smaller than E_0


y
x
E_0
0
Δz
$\frac{\pi}{2\beta}$
z
At $t=0$,
$\vec{E} = \hat{a}_y E_0 \cos(\beta z)\Big|_{z=0} = \hat{a}_y E_0$
At $\frac{\pi}{2\beta} \rightarrow E_0 = 0$



$$\vec{E} = \hat{a}_y E_0 \cos(\beta z)\Big|_{z=0} = \hat{a}_y E_0$$

At $z = \frac{2\pi}{\beta} \Rightarrow \vec{E} = E_0$

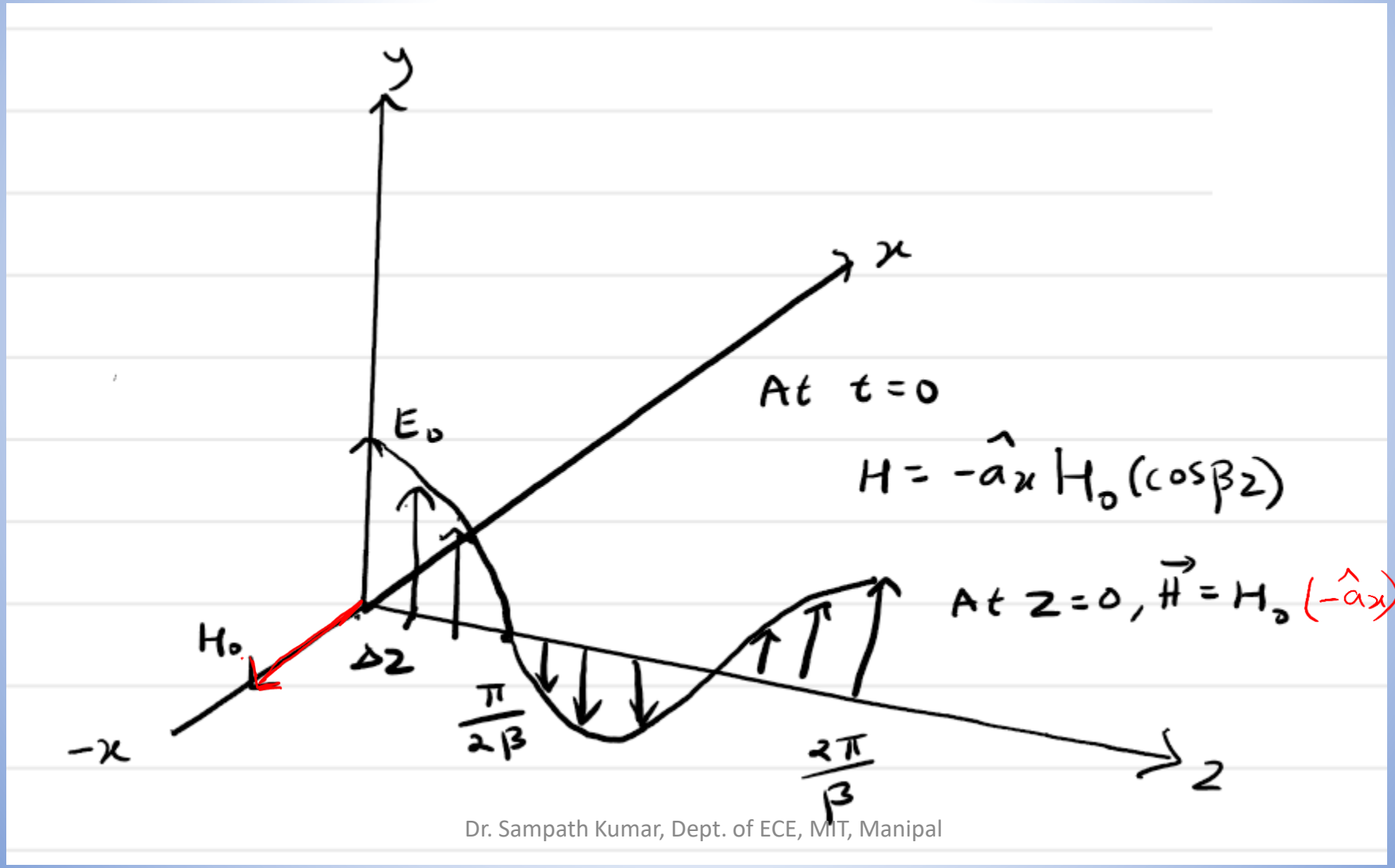
$$\lambda = \frac{2\pi}{\beta}$$

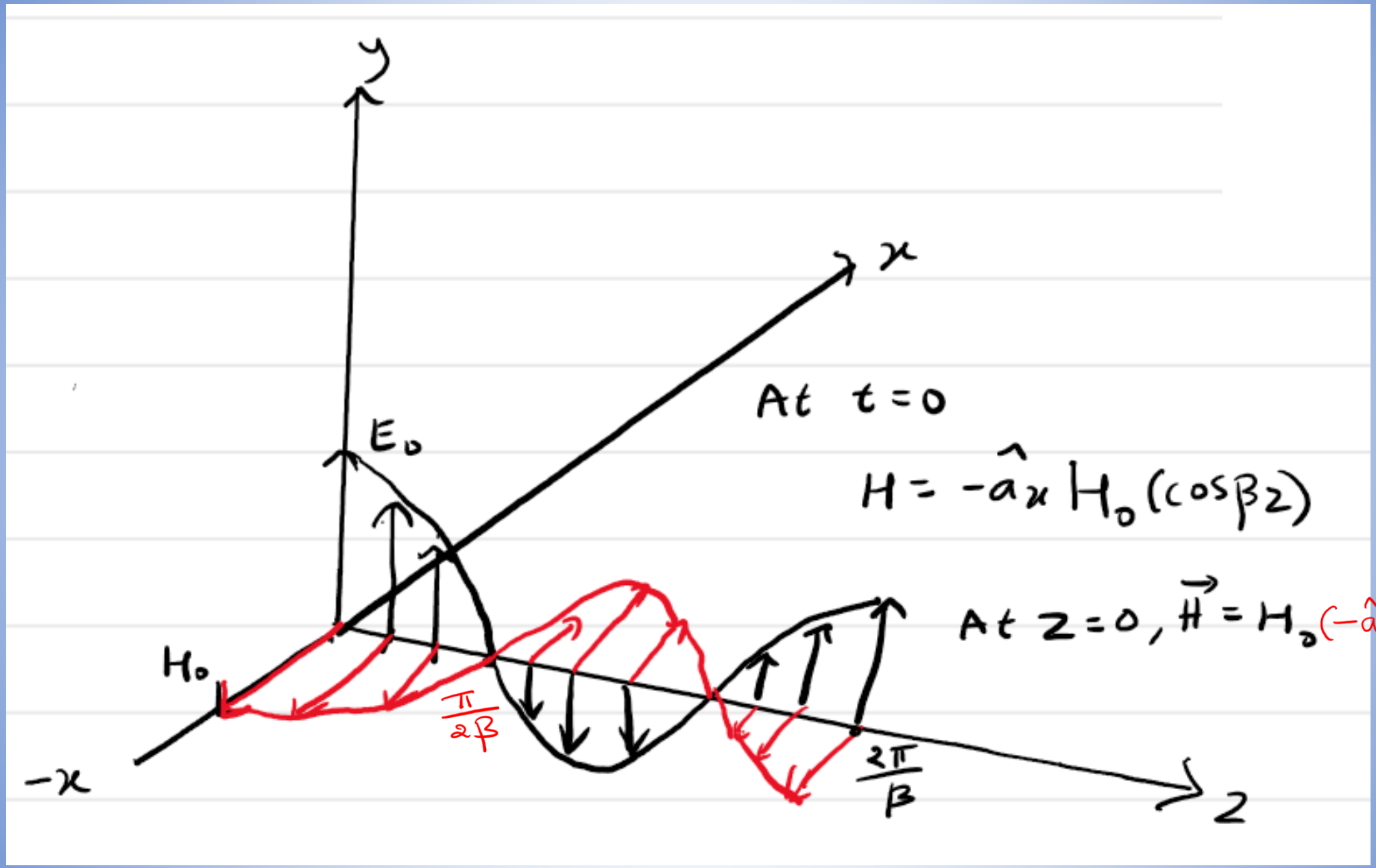
wavelength

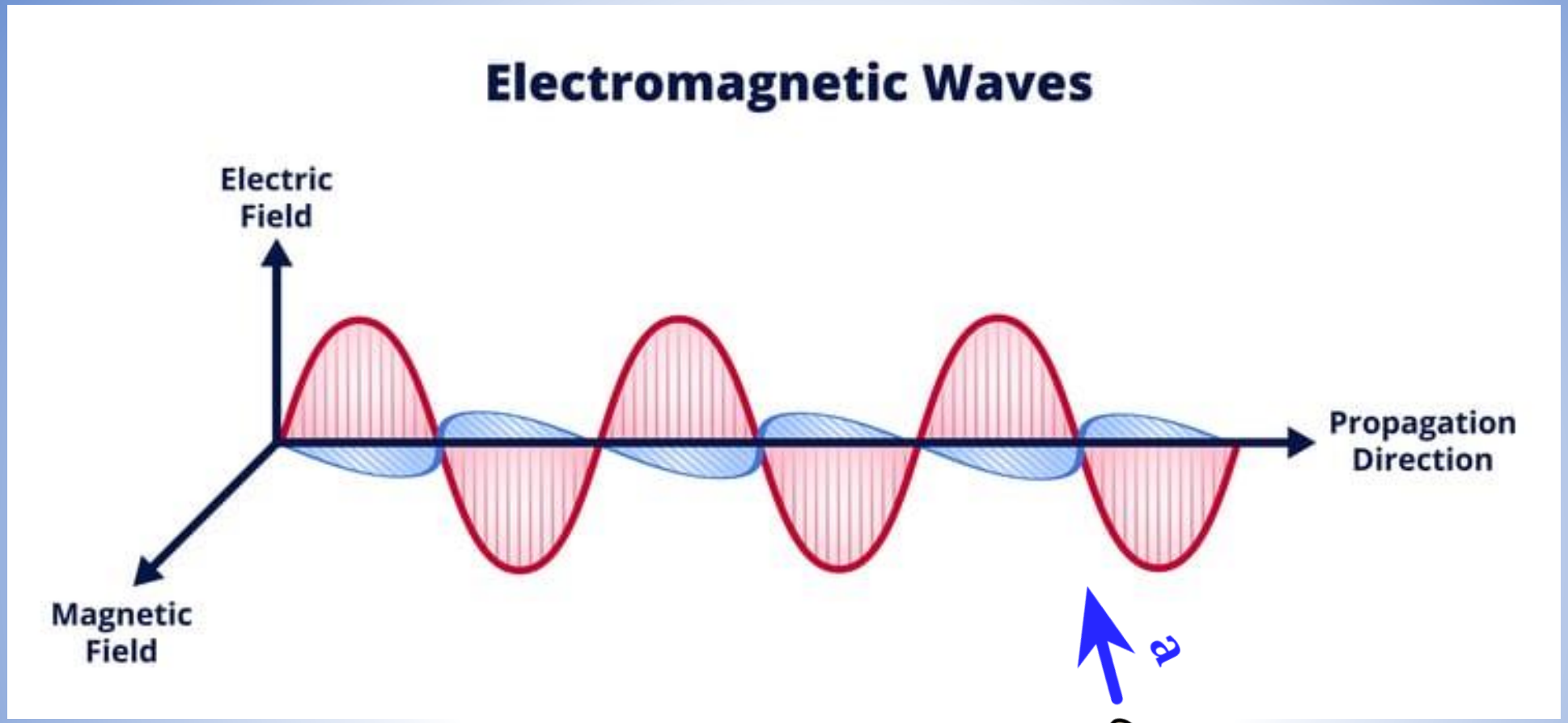
$$\beta = \frac{2\pi}{\lambda}$$

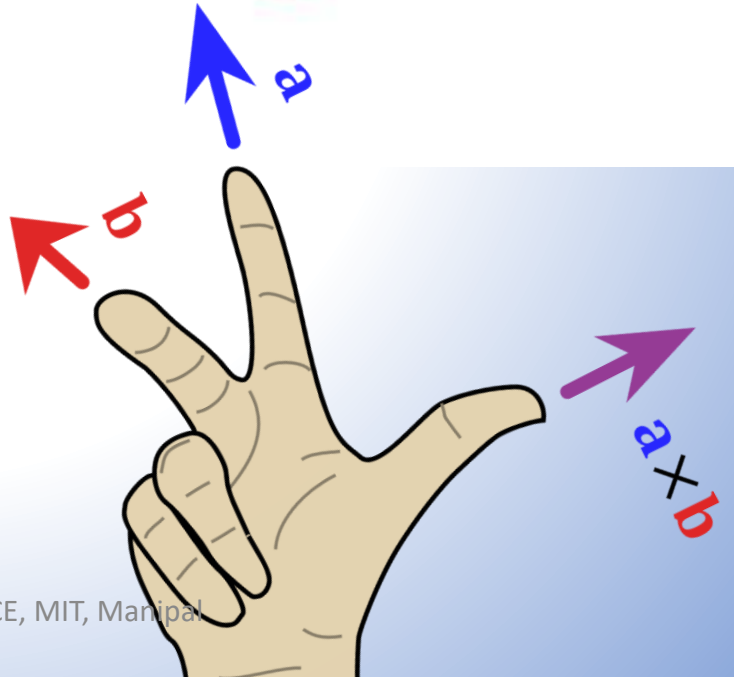
$$H = -\hat{a}_x \frac{E_0 \beta}{\omega \mu_0} \cos(\omega t - \beta z)$$

Let $\frac{E_0 \beta}{\omega \mu_0} = H_0 \Rightarrow \vec{H} = -\hat{a}_x H_0 \cos(\omega t - \beta z)$

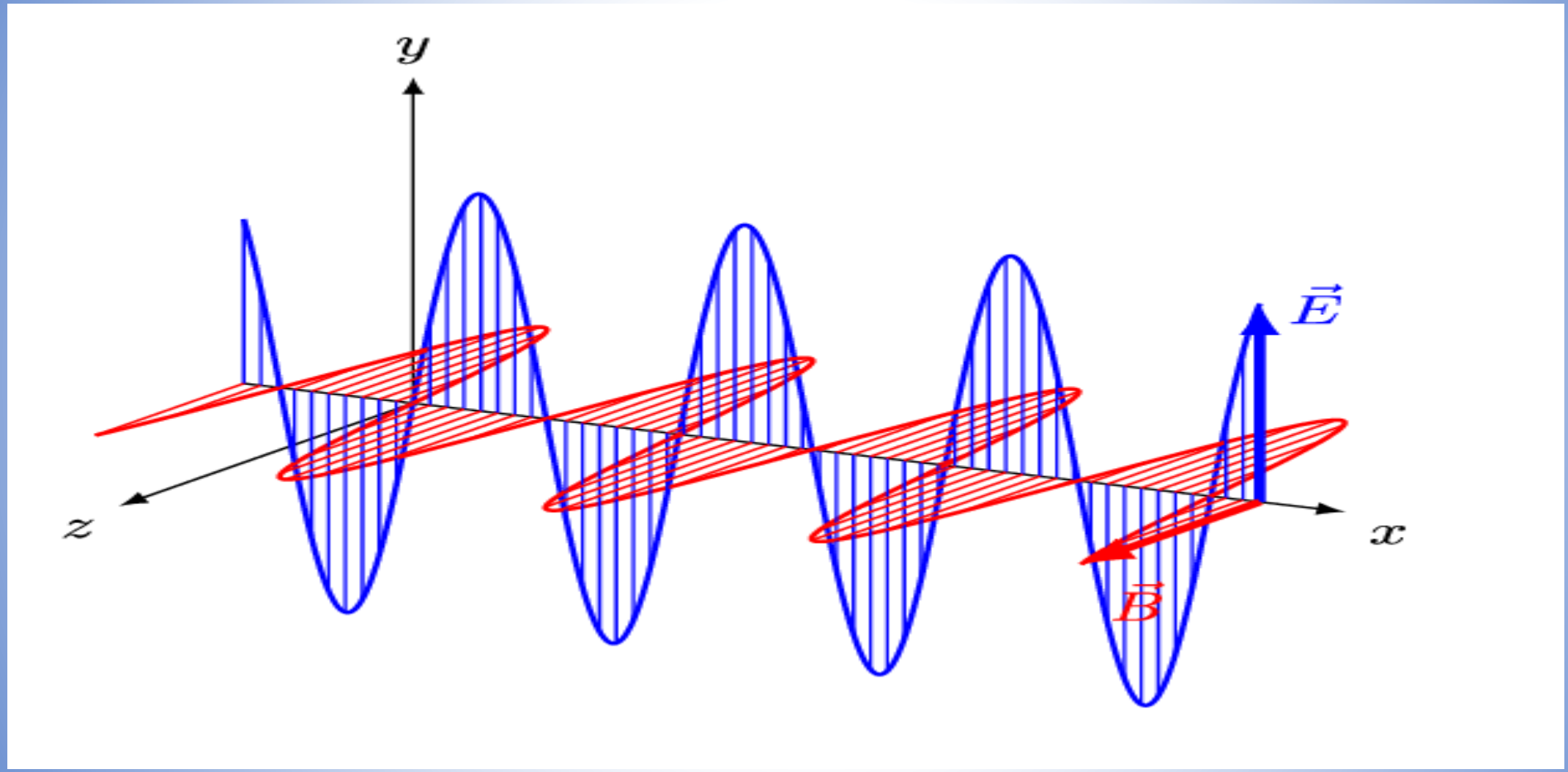



y
x
E_0
H_0
$\frac{\pi}{2\beta}$
$\frac{2\pi}{\beta}$
-x
z
At t = 0
$H = -\hat{a}_x H_0 (\cos\beta z)$
At z = 0, $\vec{H} = H_0(-\hat{a}_x)$



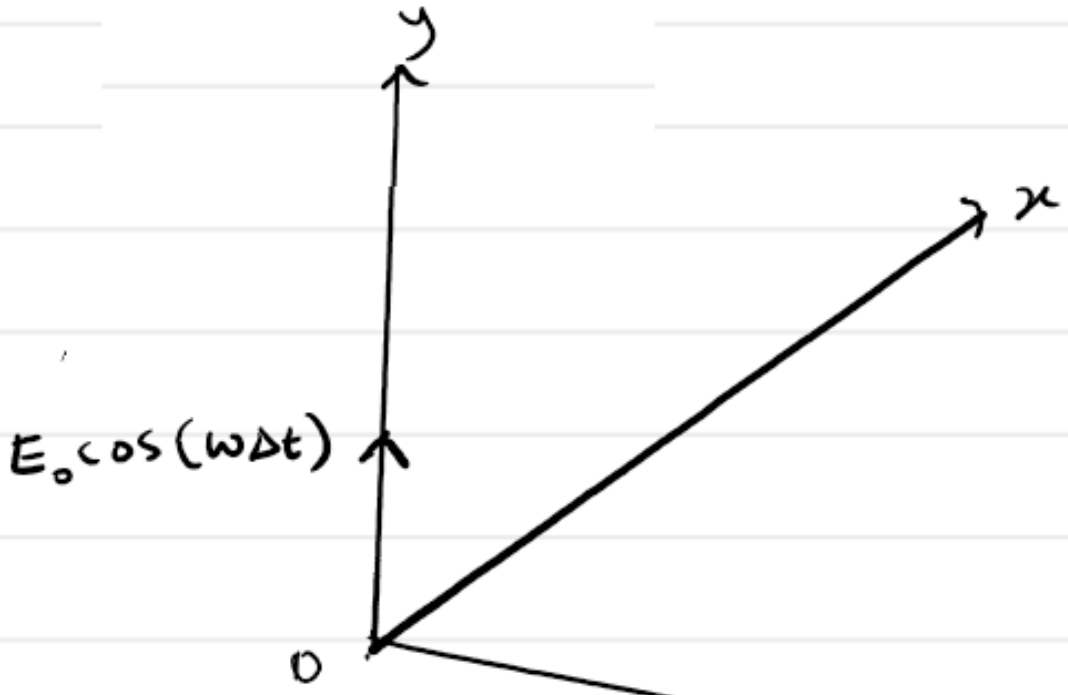


- E x H gives direction of propagation


y
z
x
$\vec{E}$
$\vec{B}$

We have plotted E and H at t = 0

Now plot these fields at t = Δt



At $t = \Delta t$

$$\vec{E} = \hat{a}_y E_0 \cos(\omega \Delta t - \beta z)\Big|_{z=0} \Rightarrow \hat{a}_y E_0 \cos(\omega \Delta t)$$

smaller than E_0

At $t=\Delta t$ and $z=\Delta z$

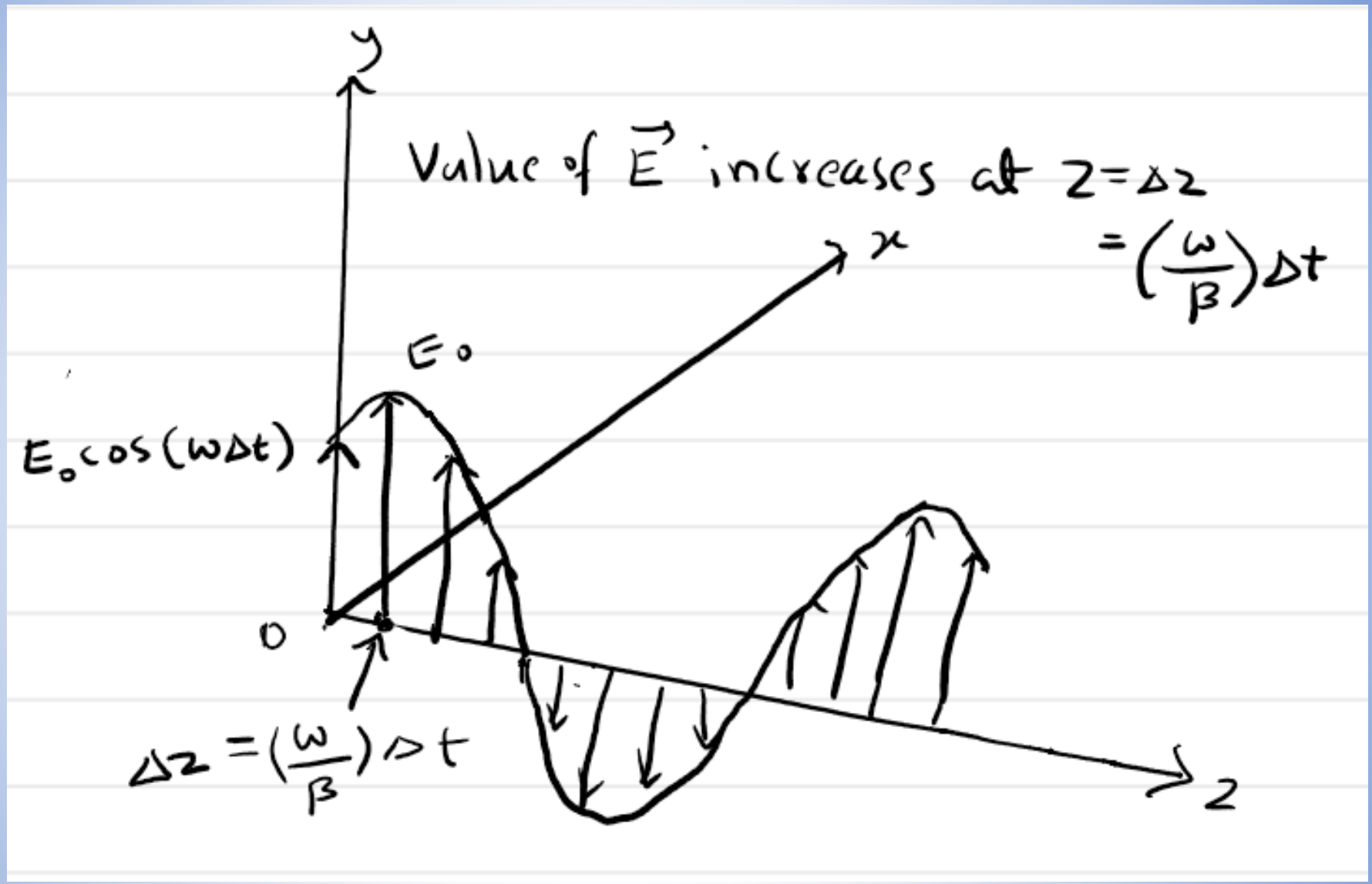
$$\vec{E}=\hat{a}_y E_0 \cos(\omega\Delta t-\beta\Delta z)$$

Let $\Delta z=\left(\frac{\omega}{\beta}\right)\Delta t$

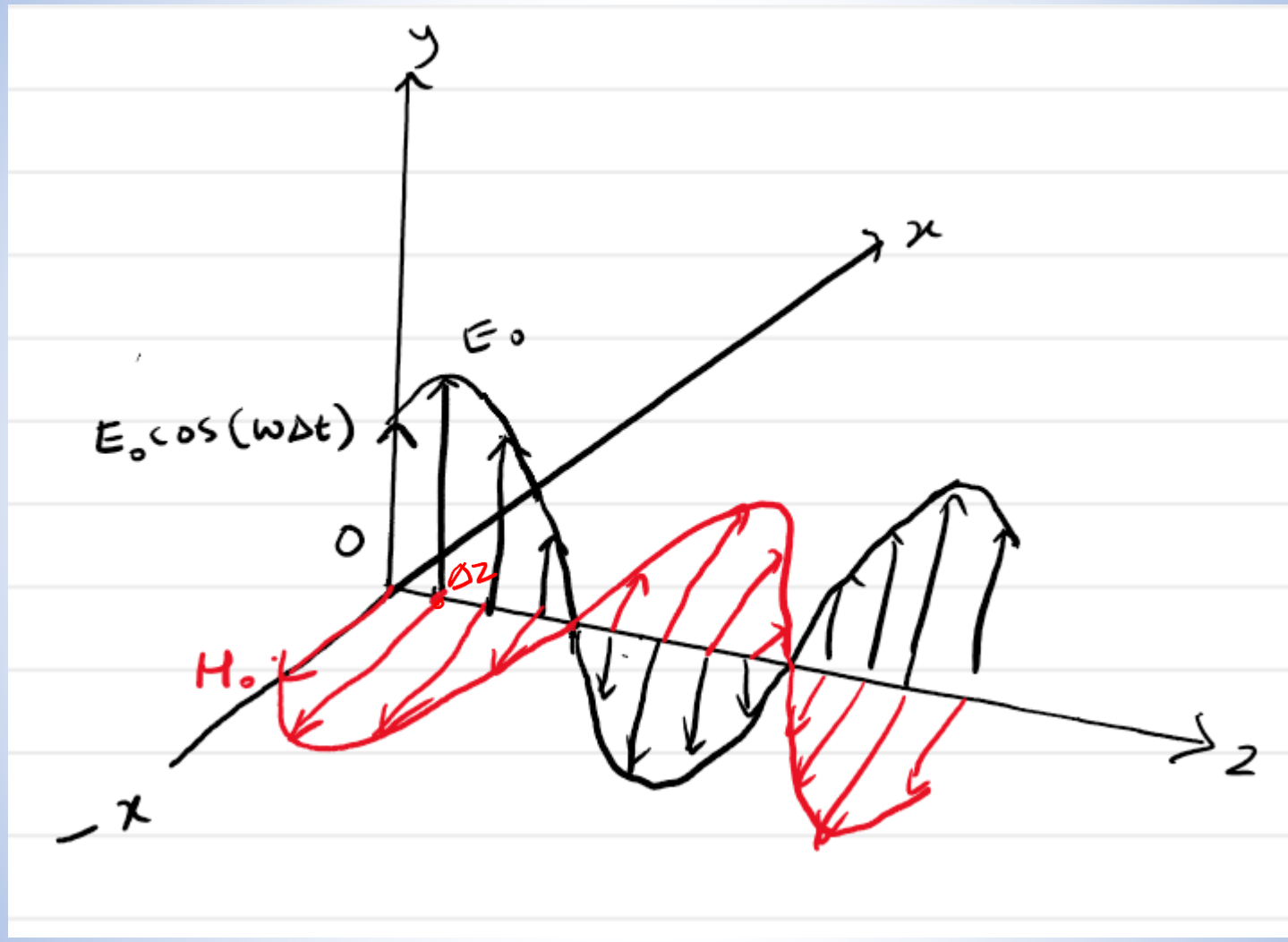
$$\Rightarrow \vec{E}=\hat{a}_y E_0 \cos\left(\omega\Delta t-\beta\left(\frac{\omega}{\beta}\right)\Delta t\right)$$

$$\vec{E}=\hat{a}_y E_0$$

Value of $\vec{E}$ increases at $z=\Delta z$


y
Value of $\vec{E}$ increases at $z=\Delta z$
$=\left(\frac{\omega}{\beta}\right)\Delta t$
x
E_0
$E_0\cos(\omega\Delta t)$
0
$\Delta z=\left(\frac{\omega}{\beta}\right)\Delta t$
z

H filed also changes in the similar manner



- As z = Δz the wave has moved in positive z direction – hence the proof.

$$\Delta z = \frac{\omega}{\beta} \cdot \Delta t$$

$$\frac{\Delta z}{\Delta t} = \text{velocity} = \frac{\omega}{\beta}$$

But $\beta = \frac{2\pi}{\lambda}$

$$\therefore \text{velocity} = \frac{\omega \cdot \lambda}{2\pi} = \frac{2\pi f \lambda}{2\pi} = f \cdot \lambda$$

= frequency x wavelength

$\therefore \vec{E}$ & $\vec{H}$ form a wave propagating in z-direction

$$H = -\hat{a}_x \frac{E_0 \beta}{\omega \mu_0} \cos(\omega t - \beta z)$$

- Take curl of H

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\text{curl } \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

$$\nabla \times H = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\frac{E_0 \beta}{\omega \mu_0} \cos(\omega t - \beta z) & 0 & 0 \end{vmatrix} = -\hat{a}_y \frac{\beta^2 E_0}{\omega \mu_0} \sin(\omega t - \beta z)$$

$$= J + \frac{\partial D}{\partial t} \quad \text{(Free space, } J = 0\text{)}$$

$$= \epsilon_0 \frac{\partial E}{\partial t}$$

Integrating both sides

$$\epsilon_0 E = \hat{a}_y \frac{\beta^2 E_0}{\omega^2 \mu_0} \cos(\omega t - \beta z)$$

But $E = \hat{a}_y E_0 \cos(\omega t - \beta z)$

Comparing

$$\epsilon_0 = \frac{\beta^2}{\omega^2 \mu_0}$$

$$\epsilon_0 = \frac{\beta^2}{\omega^2 \mu_0}$$

$$\Rightarrow \beta = \omega \sqrt{\mu_0 \epsilon_0}$$

But velocity $= \frac{\omega}{\beta}$

$$= \frac{\omega}{\omega \sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

$= c$

= velocity of light

= velocity of EMW

$$H = -\hat{a}_x \frac{E_0 \beta}{\omega \mu_0} \cos(\omega t - \beta z)$$

$$H_0 = \frac{E_0 \beta}{\omega \mu_0}$$

$$\frac{E_0}{H_0} = \frac{\omega \mu_0}{\beta}$$

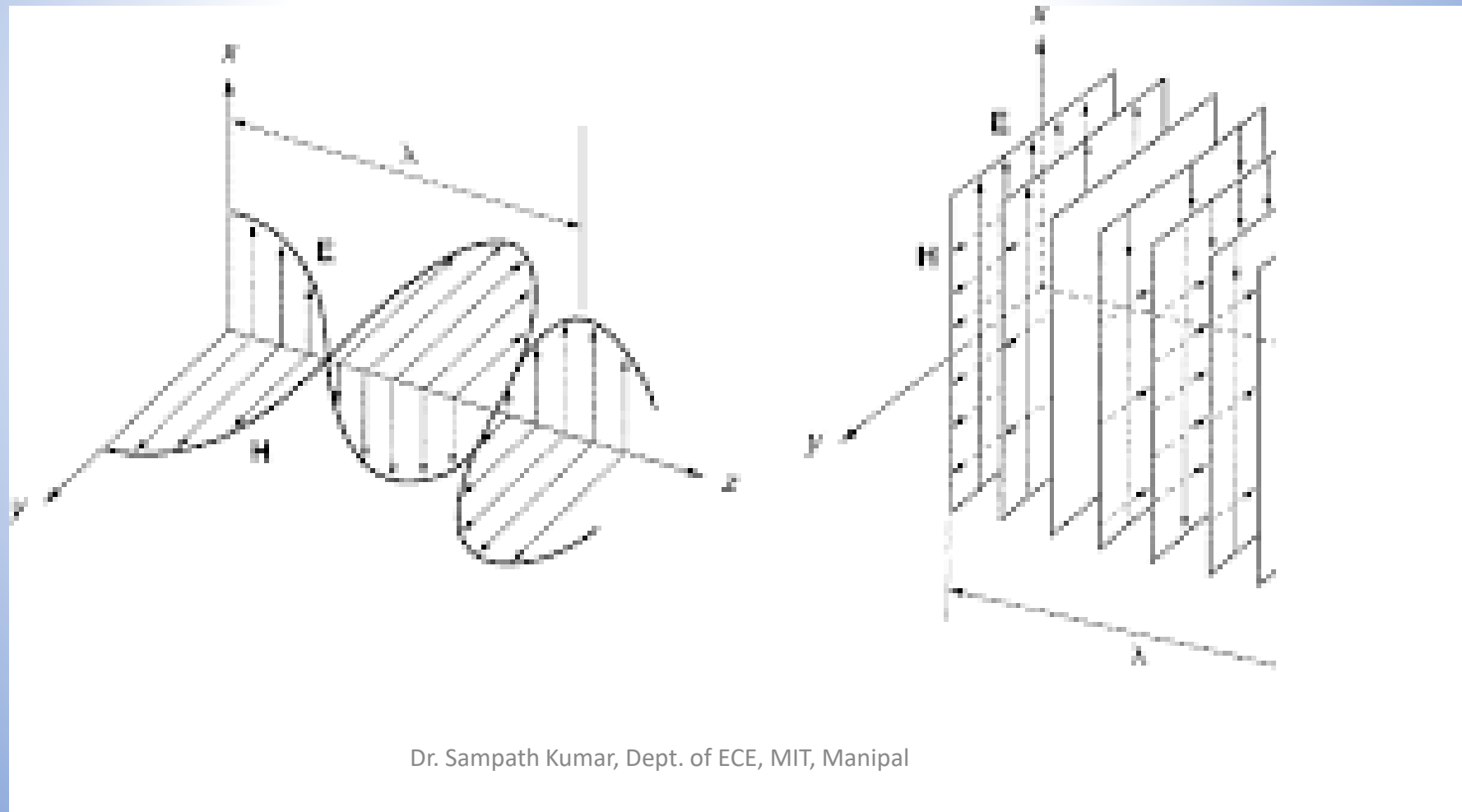
But $\frac{\omega}{\beta} = C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$\Rightarrow \frac{E_0}{H_0} = \frac{\mu_0}{\sqrt{\mu_0 \epsilon_0}} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377\ \Omega\ (120\pi\ \Omega)$$

η = Intrinsic impedance of wave in free space

Uniform plane waves

- Planes of uniform fields propagating in the form of a wave



Problem 10.1 : The magnetic field intensity of a uniform plane wave in air is 20 A / m in the $\bar{a}_y$ direction. The wave is propagating in the $\bar{a}_z$ direction at a frequency of 2 Grad / s (2×10^9 rad/s). Find : (a) the wavelength ; (b) the frequency ; (c) the period ; (d) the amplitude of $\bar{E}$.

$$\beta = \omega\sqrt{\mu_0 \epsilon_0} = 6.6712 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = 0.942 \text{ m}$$

$$f = \frac{\omega}{2\pi} = \frac{2 \times 10^9}{2\pi} = 318.3 \text{ MHz}$$

$$T = \frac{1}{f} = 3.142 \text{ s}$$

$$\eta = \frac{E}{H} = 377\,\Omega \Rightarrow \vec{E} = \eta \cdot \vec{H} = 377 \times 20 = 7540 \text{ V/m}$$


Thank
you